MIAN MUHAMMAD ABUBAKAR

Software Engineer • Full-Stack & AI Integration

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SUMMARY

Software Engineer with 4+ years of experience building production full-stack and AI-powered SaaS platforms, including a multi-tenant recruitment platform serving 10,000+ monthly active users and the UAE government's Mawaheb talent platform, built as sole full-stack developer. Owns delivery end to end, from database schema and API design to frontend and cloud deployment on AWS and Azure. Hands-on production work in multi-agent LLM orchestration, RAG, vector search, and real-time systems.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, PHP, SQL
- **Frontend:** React, Next.js, Vue.js / Nuxt.js, React Native (Expo), Tailwind, shadcn/ui, ReactFlow; TanStack Query, Redux, Zustand
- **Backend:** Node.js, NestJS, Express.js, FastAPI, Laravel, REST APIs, GraphQL, WebSockets, Server-Sent Events
- **AI / ML:** OpenAI APIs (function calling), multi-agent orchestration, RAG, embeddings (text-embedding-3-large), pgvector, real-time AI avatars (HeyGen, Anam CARA-3 over WebRTC), Affinda, custom Python ML services
- **Databases:** PostgreSQL (query optimization, indexing, schema design), Redis, MySQL, MongoDB, SQL Server, pgvector, Typesense
- **Cloud & DevOps:** AWS (EC2, S3, SES, Transcribe, Polly), Azure (Blob Storage, Key Vault, Communication Services, Speech, VMs), Docker, GitHub Actions CI/CD, OSRM self-hosted, OpenStreetMap
- **Architecture:** Microservices, Turborepo monorepos, background job queues (Bull/Redis), multi-tenant SaaS, OAuth2, Entra ID SSO
- **Testing & Tools:** Git, Jest, Vitest, PHPUnit, Stripe, Mapbox, Twilio, Pusher, Socket.io, Puppeteer

PROFESSIONAL EXPERIENCE

Software Engineer — 09/2025 - 10/2026

Aslase | Lahore, Pakistan

Mawaheb – DGE Talent & Career Platform (UAE Government, live)

- Sole full-stack developer on an AI-powered talent and career platform for the UAE's Department of Government Enablement (DGE), spanning 13 use cases. Delivered the core modules end to end: Talent Assessment with AI-generated questionnaires and async media-answer evaluation; AI Interview with real-time avatar interviewers (Anam CARA-3 over WebRTC), Azure Speech-to-Text, and a turn-based conversational interview engine; and a Skills Observatory dashboard comparing workforce demand and supply across emirates, industries, and roles. Designed a secure ML integration layer where the backend proxies every ML call (WebSocket + REST), so no credential reaches the browser, with configuration from Azure Key Vault and short-lived OAuth2 client-credential tokens.
- **Tech:** Vue 3, TypeScript, Vite, Express, Knex + Objection.js (Clean Architecture), PostgreSQL on Azure, WebSocket proxies, Azure Key Vault, Blob Storage, Communication Services, Speech SDK, Anam CARA-3 (WebRTC), MSAL / Entra ID SSO, JWT, Chart.js
- **Impact:** Replaced manual assessment and interview processes with one bilingual (Arabic/English) AI platform, keeping all data and ML credentials in a UAE-hosted, Key Vault-secured backend.

AI Employee Agent

- Multi-agent productivity platform unifying Gmail, Jira, GitHub, Notion, Google Drive, Calendar, and WhatsApp into one intelligent assistant. Built a multi-agent orchestration layer that auto-detects complex tasks and routes them to role-specific agents, plus a ReactFlow visual workflow editor for chaining steps. Implemented RAG over uploaded documents (PDF/DOCX/TXT): sentence-based chunking, text-embedding-3-large embeddings stored in PostgreSQL with pgvector, cosine-similarity top-k retrieval scoped per user, and source citations in answers. Scheduled analyzers monitor stale PRs, Jira blockers, calendar conflicts, and urgent emails.
- **Tech:** NestJS, TypeORM, React (Vite), PostgreSQL with pgvector, OpenAI function calling, text-embedding-3-large, OAuth (Google, GitHub, Jira, Notion), whatsapp-web.js, ReactFlow, Tailwind, shadcn/ui
- **Impact:** Unified 7 work apps into a single assistant, automating routine coordination (email triage, ticket creation, meeting prep, status alerts) that previously meant switching between separate tools.

Learning Management System (LMS)

- AI-powered LMS for adult professional development with 27 modular NestJS services and a Vue 3 frontend. Supports students, faculty, and leadership with adaptive learning, mock interviews powered by HeyGen avatars and OpenAI, AI career counseling, ML-driven course generation, skill gap analysis, certificate generation with PDFKit, workforce analytics, Redis caching of AI recommendations, and full multi-tenant isolation.
- **Tech:** NestJS, Vue 3, TypeScript, PostgreSQL, TypeORM, Redis, Azure Blob Storage, HeyGen API, OpenAI, ApexCharts, Chart.js, ECharts, JWT
- **Impact:** Replaced human-led mentoring and mock interviews with on-demand AI tutoring and career counseling, so client organizations can scale workforce upskilling without adding trainers.

NEP – National Experts Program (UAE Government)

- Talent assessment and career-alignment platform for the UAE's National Experts Program, evaluating candidates across AI expertise, leadership capacity, strategic thinking, growth mindset, and ethical leadership. Pipeline runs from resume upload

(Affinda + custom ML server), through a 5-question AI text interview, two phased MCQ assessments, and a final consolidated leadership score weighted across four pillars. Bull jobs handle ML polling while Socket.io pushes real-time progress.

- **Tech:** Express.js, TypeScript, PostgreSQL on Azure, Knex, Objection.js, Bull, Redis, Socket.io, OpenAI, Affinda, custom Python ML server, Azure Blob Storage, Azure Communication Services, Puppeteer
- **Impact:** Replaced weeks of manual screening for the UAE's national AI talent program with an automated multi-stage pipeline that scores every candidate against the same weighted four-pillar rubric.

Assessment & Recruitment Portal

- Multi-tenant HR and recruitment SaaS that streamlines the full hiring funnel with AI-powered candidate screening, auto-evaluated English proficiency and behavioral assessments (MCQ, speaking, writing, reading, listening with in-browser audio), GPT-based interview evaluation, video transcription, Affinda CV parsing, and intelligent candidate ranking. Multi-tenancy enforced by company-scoped middleware; a Redis distributed lock (SET NX with TTL) ensures AI scoring runs exactly once across container replicas; prompt changes validated against a regression set of test CVs.
- **Tech:** Vue 2, Vuetify, Vuex, Express.js, TypeScript, PostgreSQL, Objection.js, Knex, Bull, Redis, Socket.io, Azure Blob Storage, Azure Speech (STT), Affinda, OpenAI, AWS S3, JWT (Google & LinkedIn OAuth)
- **Impact:** Powers end-to-end hiring for 10,000+ monthly active users across multiple organizations, automating CV screening, video interview evaluation, and candidate ranking that previously consumed days of recruiter time per applicant; cut candidate evaluation turnaround from days to minutes.

Backend Test Suites

- Wrote Jest unit and integration suites for NestJS backends at Aslase: a dairy management system (~300 cases) and an industrial flange management system (~110 cases).

Full-Stack Developer **02/2025 - 08/2025**

CFIT | Lahore, Pakistan

Transportation Platform (UK Multi-Tenant SaaS)

- Architected a Turborepo monorepo with 5 applications: a Transport Management System, Warehouse Management System, and dedicated React Native mobile apps for drivers, technicians, and customers, shipped to the App Store and Google Play. Built comprehensive job management, automated Stripe invoicing, a finance system with PDF generation, and a Workshop Management System with offline-first mobile inspections. Fleet management includes real-time tracking and 10+ third-party integrations.
- **Tech:** Next.js, React, TypeScript, Convex BaaS, Drizzle ORM, PostgreSQL, React Native (Expo), React Navigation, Turborepo, TanStack Query, Redux, Stripe, Mapbox, Jest
- **Impact:** Achieved 99% uptime and 35% faster API response times, replacing several disconnected legacy tools with a unified operational platform for UK transport operators.

UK Taxi Management System

- End-to-end taxi platform featuring an intelligent dynamic pricing algorithm, an AI-powered dispatch system that optimizes driver assignment, and a secure payment layer integrating Stripe and Wonderful Payments with PCI-DSS compliance. Added real-time driver tracking, automated email notifications, and multi-app frontends for operators, drivers, and customers.
- **Tech:** PHP, Laravel, React.js, Vue.js, Next.js, Nuxt.js, TypeScript, MySQL, Redis, Stripe API
- **Impact:** 20% revenue increase from intelligent pricing and 30% shorter rider wait times via AI dispatch.

Self-Hosted OSRM Routing Engine + Typesense Address Search

- Replaced costly Google Maps API calls in the UK Taxi Management System by deploying an OSRM (Open Source Routing Machine) server on AWS EC2 that computes distances between any two points using lat/long. Imported the full UK region from OpenStreetMap, wrote Python tooling to convert OSM data into JSON, and ingested it into a Typesense addresses collection to power fast in-app address autocomplete.
- **Tech:** AWS EC2, OSRM, OpenStreetMap, Python, Typesense, Linux, Nginx
- **Impact:** Eliminated recurring Google Maps API costs and gave the platform full control over routing data, while sub-100ms Typesense autocomplete sharply improved address search for drivers and customers.

Full-Stack Developer **11/2022 - 01/2025**

SA Systems | Lahore, Pakistan

Enterprise ERP System

- Multi-tenant ERP serving 5 housing societies with 3,000+ residents and 1,000+ employees. Built 12 integrated modules including HRMS, a 5-level Chart of Accounts with double-entry ledger, a full procurement cycle (demand → quotation → PO → GRN → invoice → payment), inventory management with FIFO valuation and barcode support, project management, and real estate units management, with real-time updates via Socket.io and Pusher.
- **Tech:** Laravel, Vue.js, SQL Server (multi-DB), OAuth2 with Passport, Spatie Permission, Socket.io, Laravel Echo, Pusher, DomPDF, Maatwebsite Excel
- **Impact:** Reduced manual work by 70% across daily operations for housing societies, replacing fragmented spreadsheets and paper processes with a single source of truth.

Triangle POS (Automotive Shops)

- Modular point-of-sale system tailored for automotive service shops and retail businesses. Built real-time Livewire-driven checkout with cart management, tax/discount handling, and partial payments, alongside inventory tracking with barcode support, stock

alerts, and multi-unit pricing. Includes 6 reports and Twilio WhatsApp/SMS service reminders.

- **Tech:** PHP, Laravel, Livewire, MySQL, Bootstrap with CoreUI, Spatie Permission and MediaLibrary, DomPDF, Maatwebsite Excel, Twilio SDK, Milon Barcode, PHPUnit
- **Impact:** Replaced paper-based billing and manual stock tracking, giving shop owners real-time visibility into sales, inventory, and customer service reminders from a single dashboard.

HR & Payroll Management System

- End-to-end HRMS designed for organizations with 1,000+ employees. Features biometric attendance, leave accruals, full payroll (overtime, deductions, allowances), loans and final settlements, and 12+ analytics dashboards.
- **Tech:** Laravel, Vue.js, SQL Server, T-SQL stored procedures, ZKTeco SDK for biometrics, ApexCharts, Chart.js
- **Impact:** Reduced payroll processing time by 60% and eliminated end-of-month manual reconciliation across HR and finance teams.

E-commerce Storefronts

- Shipped multiple e-commerce websites for SA Systems clients with catalogs, payments, order workflows, role-based access, and admin dashboards.
- **Tech:** Vue.js / Nuxt.js, Laravel, MySQL, Stripe, Tailwind CSS
- **Impact:** Moved small and mid-size businesses from offline-only sales to online stores with payments and order tracking.

Software Engineer Intern **08/2022 - 10/2022**

Devsinc | Lahore, Pakistan

- Trained on the MERN stack (MongoDB, Express, React, Node.js) in an Agile team under senior engineers, working with code reviews and branch-based Git workflows.
- Built a task-management module (REST API and React UI) and presented it to the engineering team at the intern demo.

EDUCATION

Master of Philosophy in Computer Science | NCBA&E, Lahore, Pakistan **2024 - 2026**

Coursework: Advanced Analysis of Algorithms, Advanced Operating Systems, Big Data Analytics, Tools & Techniques of Data Science, Advanced Machine Learning, Advanced Topics in AI.

Bachelor of Science in Computer Science | NCBA&E, Lahore, Pakistan **2019 - 2023**

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Distributed Systems, Software Engineering, Computer Networks.

LANGUAGES

English - Professional Working Proficiency | Urdu - Native / Full Professional Proficiency